AVLA PoC Final Report

Project: AVLA (Audio-Visual Localization & Analysis) PoC

Model Architecture: SimpleHead (Linear Classifier on top of WavLM 768-dim embeddings)

Training Configuration:

Max Epochs: 30
Learning Rate: 0.05
Early Stopping Patience: 3
Triggered Epoch: 16
Best Epoch Restored: 13

Performance Metrics:

Initial Loss: 0.689533
Best Validation Loss: 0.001572
Relative Loss Reduction: 99.77%
Test Accuracy: 100.0%

Inference Evaluation:

Siren events perfectly classified with high confidence (>99.8%) and background with near-zero confidence (<0.03%). Timeline successfully profiled on clip C01.